ITCS224: Fundamental of Information Systems & Software Development

Faculty of Information and Communication Technology

Project Phase 2 of Semester 2/2025

Score: 25 points

Submission Date:

Saturday 18th April 2026 @ 20:00 via MyCourses

Presentation date (Slides + Live demo + Q&A):

Section 1: 23 April 2026 @ 13.30 – 16.30
Section 2: 24 April 2026 @ 09.00 – 12.00
Section 3: 21 April 2026 @ 13.00 – 16.00

Description

This project simulates a software development process. Each group acts as a software development company. You will develop a system or application to support the business processes defined in Project Phase 1. The project has four sub phases (SPs): business and system definition, requirements analysis, detailed analysis and design diagrams, and prototyping and testing. The main goal is to help you keep your analysis and design consistent across all artifacts. After you identify all use cases, your group must select three main use cases that reflect clear actor-based scenarios. You will use these three selected use cases as the basis for the use case narratives, DFD Level 2, the prototype, and the test cases. The details are as follows.

SP1: Business and system definition

In this sub-phase, students must define their business and identify key processes that require software support. You must provide a brief business overview and describe the high-level purpose and goals of the system or application you intend to develop. In this phase, you may revise your original idea to focus more on a clear software system description, so the later steps are clearer, i.e., making it easier to identify functional requirements.

SP2: Requirements analysis

In this sub phase, students must identify and document the functional and non functional requirements of the system. Each requirement must be clearly stated and must have a unique identifier (e.g., R1, R2) to support traceability. In addition, you must develop a complete use case diagram that covers the entire system scope. From all identified use cases, your group must select three main use cases and write the use case narratives for these selected use cases.

SP3: Detailed analysis and design diagrams

In this sub phase, students must create Data Flow Diagrams (DFDs) to show the system in detail, with accuracy and consistency across all levels. These diagrams should show your understanding of software modeling and how data moves between actors, processes, and data stores. The required DFD set includes DFD Level 0 (Context Diagram), DFD Level 1 covering all main processes, and DFD Level 2 for the processes that relate to the three selected use cases.

SP4: Prototyping and testing

In this sub-phase, students are required to build a prototype of their system or application that aligns with their analysis and design. You may use vibe coding tools to design your prototype (e.g., Canva Code), or you may implement the prototype with code from scratch. The prototype must show how users can perform the system functions based on your design. The prototype scope must cover the three selected use cases from SP2, and your group must create test cases to verify these three use cases on the prototype, including expected results for both successful and unsuccessful scenarios.

Team members: 5 - 6 students/group, as in Project Phase#1.

Deliverables: ****Please name all your files in English****

No.	Item	Minimum expectation	Score (%)
1	Introduction of the work (e.g., overview of your business, description of your proposed system and your end-users)	500 words	5

No.	Item	Minimum expectation	Score (%)
2	Functional requirements and non-functional requirements	At least 20 functional requirements and 10 non-functional requirements.	10
3	Use case modeling (Use diagram and narratives)	One use case diagram that covers all functional requirements. Then, select 3 main use cases and write 3 use case narratives (one narrative per selected use case).	10
4	Functional decomposition diagram (Structure chart)	Covers all main processes. No more than 7 processes at the top level.	5
5	Data Flow Diagram Level 0 (Context diagram)	One complete context diagram. Must align with requirements, use case diagram, and functional decomposition diagram.	5
6	Data Flow Diagram Level 1	Covers all processes. Must be consistent with DFD Level 0.	15
7	Data Flow Diagram Level 2	Create DFD Level 2 for the processes that relate to the 3 selected use cases. The Level 2 diagrams must be consistent with DFD Level 1.	15
8	Prototype	Prototype must cover the 3 selected use cases and show user flows and key screens. Must align with the DFDs.	15
9	Test cases	Test the prototype for the 3 selected use cases. Provide 2 test cases per use case: one happy path and one	15

No.	Item	Minimum expectation	Score (%)
		unsuccessful case, using the given test template.	
10	Quality of your presentation & Q&A	Demonstrate a good understanding of your project, explain all the required components and be well-prepared to answer questions.	5
Total			**100**

Detailed description

1. **Introduction of the Work**

   You must provide an overview of your business and its customers. Then, identify the business processes that require software support and explain why. You must also describe the proposed software system, its purpose, its main screens or appearance, and its intended users.

2. **Functional Requirements and Non-functional Requirements**

   You must identify at least 20 functional requirements and at least 10 non-functional requirements for your system or application. Each requirement must be clearly stated and must have a unique ID for traceability. Functional requirements describe what the system must do to support the business processes. Non-functional requirements describe quality attributes and constraints of the system, such as usability, performance, security, and reliability.

3. **Use Case Modeling (diagram and narratives)**

   You must provide a complete use case diagram that covers the system scope and supports all functional requirements. From all identified use cases, your group must select three main use cases that reflect clear actor-based scenarios. You must write three use case narratives, one for each selected use case, using the given template. Each

narrative must include the use case ID, title, actors, preconditions if any, main flow, alternative or unsuccessful flow, and postconditions if any.

4. **Functional Decomposition Diagram**

   You must create a functional decomposition diagram that covers all main processes and all essential functional requirements. The diagram must have no more than 7 processes at the top level. The decomposition may include more than two levels when needed.

5. **Data Flow Diagram Level 0 (Context Diagram)**

   You must submit a complete context diagram for the whole system. It must show the system boundary, external entities, and major data flows. It must align with the requirements, use case diagram, and functional decomposition diagram.

6. **Data Flow Diagram Level 1**

   You must submit a Level 1 DFD that covers all main processes. It must be consistent with the context diagram and the functional decomposition diagram.

7. **Data Flow Diagram Level 2**

   You must create DFD Level 2 for the processes that relate to the three selected use cases. These Level 2 diagrams must provide enough detail to explain how each process works and must align with DFD Level 1. The selected use cases and processes must focus on primary user activities. Do not select account creation or authentication processes, such as create account, log in, or reset password.

8. **Prototype**

   Based on the three selected use cases and their related Level 2 DFDs, you must create a prototype of your system. The prototype must show key screens, user interactions, and page flows for completing the three selected use cases. In the report, include screenshots and describe each screen. You must also explain step by step how a user navigates the prototype to complete each selected use case. In addition, you must provide the URL of your prototype and ensure that it is accessible.

9. Test cases

You must develop test cases to test the prototype for the three selected use cases. Provide two test cases per selected use case, one happy path and one unsuccessful case, using the techniques discussed in class and the provided template. Each test case must include a test ID, test name, test procedure, and expected result. The procedure must describe the steps on the prototype and the expected output or behavior.

Impact factors of your score are:

- Completeness: Your analysis should be thorough. This includes ensuring that lists of functional and non-functional requirements are comprehensive, and all relevant actors are identified.
- Diagram Accuracy: All diagrams should adhere to the correct standards and effectively convey their intended meaning.
- Consistency: Your analysis and diagrams should align seamlessly. Actors and processes depicted must match the business description and requirement specifications.
- Deliverable Quality: Consider factors such as readability, spelling, and grammar when preparing your documents.
- Engagement Quality: Your responsiveness and the quality of your answers during Q&A sessions will also be evaluated.

Presentation Criteria:

1) The presentation must include at least all of the components mentioned earlier.
2) The length of the presentation **cannot exceed 15 minutes**.
3) The **Q&A session is 10 minutes**. The questions will be asked by the lecturer.
4) Please ensure that all the diagrams are clear enough to be easily read.
5) You also need to present other necessary parts to make your presentation understandable and complete.
6) All members must be present for the presentation and each member must give a presentation.
7) Each group must come on time, if your group cannot present for your presentation slot, the score will be deducted.

Communication:

1) If you have any questions regarding the clarification of the projects, you can post them in Microsoft Teams – the Discussion channel.
2) Please post one question per post.
3) When you post, please tag the lecturer.
4) Please use "reply" on your post to discuss the same question.
5) Do not send any private messages to ask any questions, we will not reply to the private messages.

Submission on MyCourses:

1) ***Report (strictly in pdf format):*** This is the report file of the topics given above with the use of the given cover template. You need to name your file as *"SecX_GroupName_BusinessName*_P2_Report" and use the given cover page.
2) ***Diagram files (strictly in pdf format):*** Please ensure that your diagrams are readable. If they are too small to be included in the report, you may add the diagram files to the submission folder. You must name your diagram files properly (e.g., DFD_L0, DFD_L1_ProcessX, DFD_L2_ProcessX.Y).
3) Put all your files into one folder then name your folder as *"SecX_GroupName_BusinessName _P2"*. After that, you need to compress this folder in the zip, rar or 7zip formats, and upload your zipped file directly on MyCourses.
4) Each group must assign one member to submit. One submission for each group only.

Your score will be deducted if you do the followings:

1. *** Only the work that is submitted before the deadline is graded. Having no work submitted before that means a ZERO score ***
2. Failure to follow the instructions will make you lose points.
3. Any issues/problems/obstacles in progressing your project must be informed before the deadline.
4. Any issues/problems that happen outside the rules stated above will be considered by all instructors. The decision and the penalty from the instructors are final.
5. In the case of plagiarism, the project phase 2 score of the plagiarizing group will be ZERO.